

Course Name: CSA0311

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Q1. Topological Sort using DFS with Cycle detection.

Sol. Algorithm:

TOPOSORT (G)

Input : Directed Graph $G(V, E)$

Output : Topological Ordering or Cycle detected.

1. Create Arrays :

visited $[V] = \text{FALSE}$

instack $[V] = \text{FALSE}$

2. Create an Empty stack S.

3. For each vertex u in V .

 If visited $[u] = \text{FALSE}$

 If DFS $(u) = \text{TRUE}$

 Print "Cycle Detected"

 Return

4. While S is not Empty

 Print Pop(S)

END.

DFS(u)

1. visited [u] = TRUE
2. instack [u] = TRUE
3. For each neighbour v of u

If visited [v] = FALSE

If DFS(v) = TRUE

Return TRUE

Else If instack [v] = TRUE

Return TRUE

4. instack [u] = FALSE

5. Push v into stack s

6. Return FALSE.

Complexity:

Time Complexity = $O(V+E)$

Space Complexity = $O(V)$

DFS is started from every unvisited vertex.

Popping the stack gives the Topological order.

Q2. Construct MST with Budget Limit, Mandatory Edges and Forbidden Edges.

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Modified Kruskal Algorithm.

Constrained - MST (G , Budget)

Input:

Graph $G(V, E)$

Mandatory Edge set M

Forbidden Edge set F

Budget B

Output:

MST or Failure

1. $MST \leftarrow \text{Empty}$
2. $\text{Cost} \leftarrow 0$
3. Initialize Union-Find structure
4. For each edge e in M
 if $\text{Find}(u) \neq \text{Find}(v)$
 Add e to MST
 Union (u, v)
 $\text{Cost} \leftarrow \text{cost} + \text{weight}(e)$

5. Remove all edges e from graph

6. Sort remaining edges in increasing order of weight

7. For each edge $e(u, v)$

If $\text{find}(u) \neq \text{find}(v)$

If $\text{cost} + \text{weight}(e) \leq \text{Budget}$

 Add e to MST

 Union (u, v)

$\text{cost} \leftarrow \text{cost} + \text{weight}(e)$

8. If Number of edges in MST $= |V| - 1$

 Print MST

 Print Total Cost

Else

 Print "MST cannot be